

Lecture 19 - Direct Current Circuits

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For solving the circuit problems many students do not know where to start. Even when they have started, they found the system to have too many variables. Actually we only need the following 2 laws in Physics to study the current in circuits:

Conservation of Charges: Whatever go into a junction must come out from the junction.

Conservation of Energy: The change in potential energy is 0 if charges come back to the start point on a loop.

Remember the term circuit - a circuit is a **closed loop** with the flow of something in it - the something here are the moving charges.

The circuits can also be largely and systematically simplified. I will make it doable for everyone.

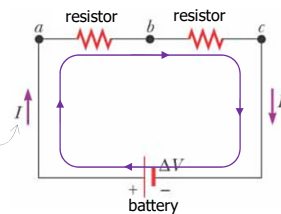
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The slides in this lecture is related to Chapter-26 of Young's textbook.

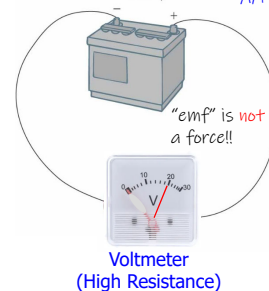
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Direct Current

- When the current in a circuit has a constant magnitude **and direction**, the current is called **direct current**
- Because the potential difference between the terminals of a battery is constant, the battery produces direct current
- The battery is known as a source to provide the **electromotive force (emf) $\mathcal{E}$** . The emf of a battery is the maximum possible voltage that the battery can provide between its terminals. The emf is measured by a voltmeter of high resistance so that the flow of current to the voltmeter is negligible



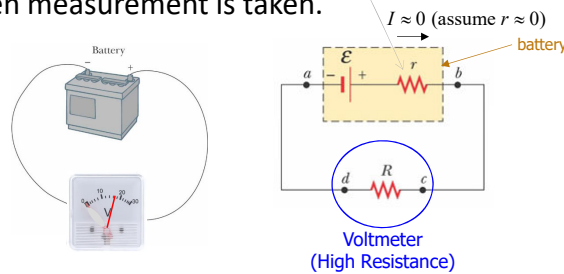
"emf" is the potential difference created by battery.
- A/Prof Tay



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Direct Current

- The internal resistance of the battery does not affect the measurement of emf because the flow of current through the internal resistance in the battery is close to 0 when measurement is taken.

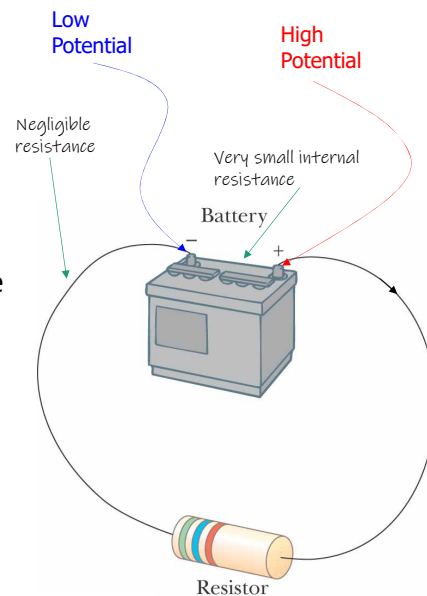


- The emf actually supplies only energy. It does not apply any force to the charges. This energy comes from the chemical energy in the battery.

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Sample Circuit

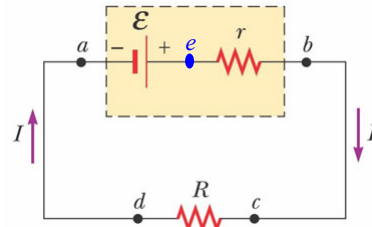
- We consider the wires in the circuit to have negligible resistance
- The positive terminal of the battery is at a **higher** potential than the negative terminal. E.g. $12\text{ V} > 0\text{ V}$; $9\text{ V} > -3\text{ V}$; $-11\text{ V} > -23\text{ V}$
- We also assume that the internal resistance of the battery is negligible unless stated otherwise



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Internal Battery Resistance

- For ideal cases the internal resistance in battery is zero so the terminal voltage ΔV (terminal voltage refers to the voltage between a and b) equals the emf $\mathcal{E}$
- But in a real battery, there is internal resistance r , and the terminal voltage is reduced to $\Delta V = \mathcal{E} - Ir$



The terminal voltage from a real battery is always smaller, and can be derived from the conservation of energy as follows:

In clockwise for 1 coulomb of charge and compare the magnitudes:

$$\Delta U_{a \text{ to } e} + \Delta U_{e \text{ to } b} + \Delta U_{b \text{ to } c} + \Delta U_{c \text{ to } d} + \Delta U_{d \text{ to } a} = 0$$

$$\mathcal{E} + (-Ir) + 0 + \frac{\Delta U_{c \text{ to } d}}{1 \text{ C}} + 0 = 0$$

$$\mathcal{E} + (-Ir) + 0 + \Delta V_{b \text{ to } a} + 0 = 0$$

$$\mathcal{E} + (-Ir) + 0 + (-\Delta V_{a \text{ to } b}) + 0 = 0$$

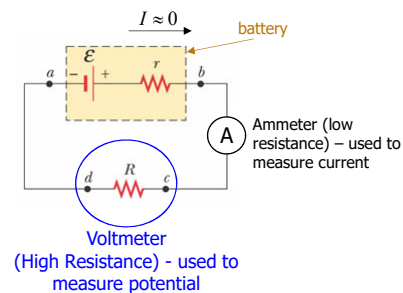
→ $\mathcal{E} - Ir = \Delta V_{a \text{ to } b}$. But $\Delta V_{a \text{ to } b}$ is the terminal voltage ΔV .

So the terminal voltage $\Delta V = \mathcal{E} - Ir$.

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EMF, cont

- The emf is equivalent to the open-circuit voltage
 - This is the terminal voltage when no current is in the circuit
 - Emf is the voltage labeled on the battery
- The actual potential difference between the terminals of the battery depends on the current in the circuit



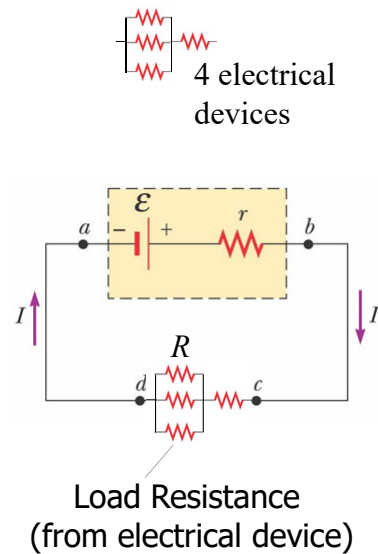
$$R = \frac{\Delta V}{I} \rightarrow \Delta V = I \times R$$

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Load Resistance

- The terminal voltage also equals the voltage across the (equivalent) external resistance

- This external resistor is called the *load resistance* (R)
- In general, the load resistance could come from any electrical device and the combinations



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$$\text{Power} = I \times \Delta V = I^2 R = \frac{\Delta V^2}{R}$$

(Power is also conserved !!!)

Power

- The total power output of the battery is $P = I \times \mathcal{E}$. This is the everything the battery can deliver.
- This total power is delivered to the internal resistor ($I^2 r$) and to the external resistor ($I^2 R$)

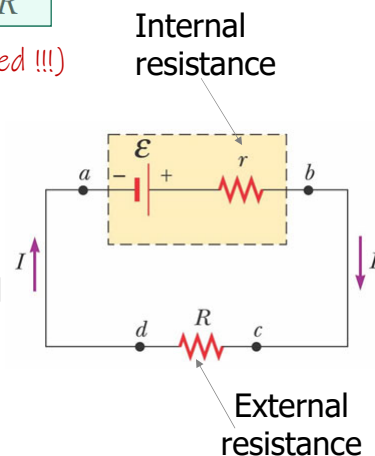
$$P = I\mathcal{E} = I^2 R + I^2 r$$

Everything the battery has

Power spent on external work

Power due to internal consumption

Likewise in life!!



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Resistors in Series



- When two or more resistors are connected end-to-end, they are said to be in *series*
- For a series combination of resistors, the currents are the same in all the resistors because the amount of charges that passes through one resistor must also pass through the other resistors in the same time interval (Conservation of Electric Charges)
- The potential difference will divide among the resistors such that the **sum** of the potential differences across the resistors is equal to the total potential difference across the **combination**

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Resistors in Series, cont

- By conservation of energy

In clockwise:

$$\Delta U_{c \text{ to } a} + \Delta U_{a \text{ to } b} + \Delta U_{b \text{ to } c} = 0$$

$$\rightarrow -\Delta U_{c \text{ to } a} = \Delta U_{a \text{ to } b} + \Delta U_{b \text{ to } c}$$

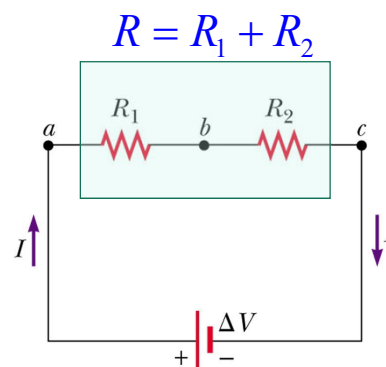
$$\rightarrow \Delta U_{a \text{ to } c} = \Delta U_{a \text{ to } b} + \Delta U_{b \text{ to } c}$$

$$\rightarrow -\Delta V = (-IR_1) + (-IR_2)$$

$$\rightarrow \Delta V = I(R_1 + R_2)$$

- Equivalent resistance R is the sum of each individual resistance in series, i.e.,

$$R = R_1 + R_2$$



(The height of a mountain is the sum of all its individual and consecutive segments!!)

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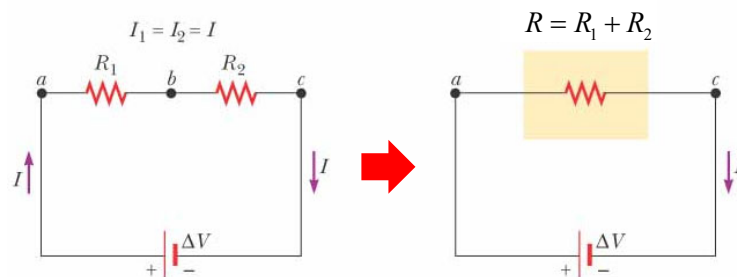
Equivalent Resistance – Series

- $R_{eq} = R_1 + R_2 + R_3 + \dots$
- The equivalent resistance of a series combination of resistors is the algebraic sum of the individual resistances and is always greater than any individual resistance
- If one device in the series circuit creates an open circuit, all other devices will be inoperative, i.e., a single point of failure will cause a complete failure in the series circuit



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Equivalent Resistance – Series – An Example



- Current is a constant in a loop
- Emf is shared by all resistors in the loop.
(Actual height of mountain is the sum of individual and consecutive segments.)
- Two resistors are replaced with their equivalent resistance R and so on

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Resistors in Parallel

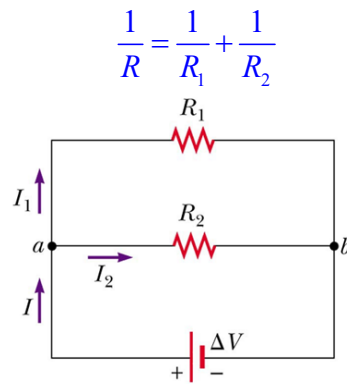
- The potential difference across each resistor is the same because each is connected directly across the battery terminals
- By conservation of electric charges, the current I that enters a junction must be equal to the total current leaving that junction, i.e.,

$$I = I_1 + I_2 = \frac{\Delta V}{R_1} + \frac{\Delta V}{R_2}$$

- Equivalent resistance R :

$$\frac{\Delta V}{R} = \Delta V \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

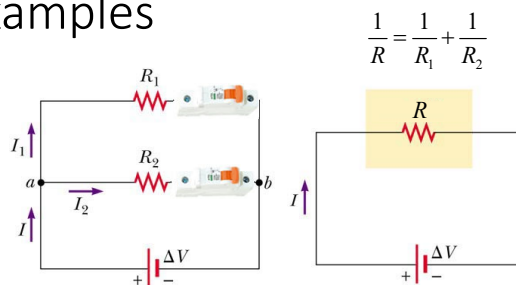
$$\Rightarrow \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R = \frac{R_1 \times R_2}{R_1 + R_2}$$



$$R = \frac{\Delta V}{I} \Rightarrow I = \frac{\Delta V}{R}$$

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Equivalent Resistance – Parallel, Examples



Circuit Breakers

- Equivalent resistance R replaces the two original resistances
- Household circuits* are wired so that electrical devices are connected in parallel
 - Circuit breakers may be used in series with other circuit elements for safety purposes

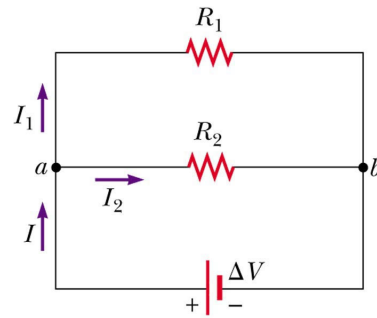
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Equivalent Resistance – Parallel

- Equivalent Resistance R

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

- The inverse of the equivalent resistance of two or more resistors connected in parallel is the algebraic sum of the inverses of the individual resistance



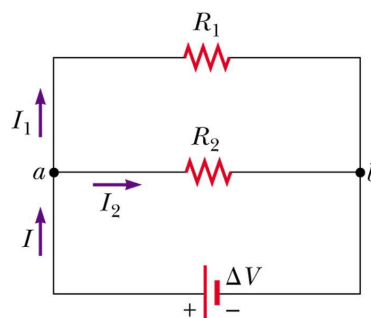
- The equivalent is always less than the smallest resistor in the group

$$\left(\frac{1}{\frac{1}{\text{smallest}} + \delta} = \frac{\text{smallest}}{1 + \delta \times \text{smallest}} < \text{smallest} \right)$$

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Resistors in Parallel, Final

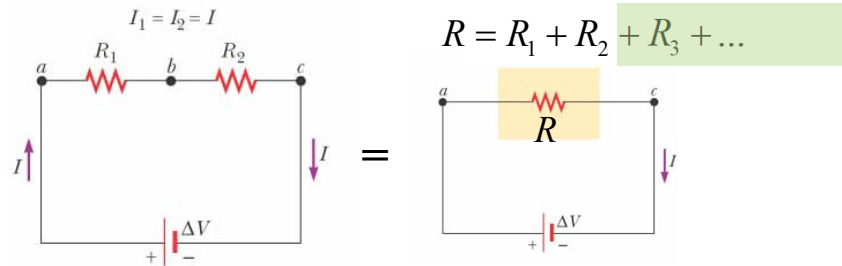
- In parallel, each device operates independently of the others so that if one is switched off, the others remain on
- In parallel, all of the devices operate on the same voltage
- The current flows on all the paths
 - The lower resistance will have higher currents
 - Even very high resistances will have some currents



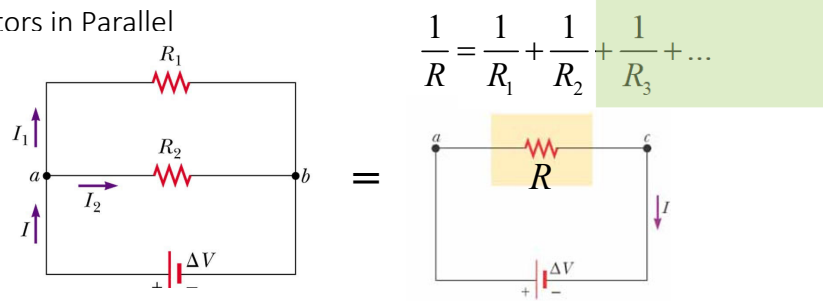
$$R = \frac{\Delta V}{I}$$

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Resistors in Series

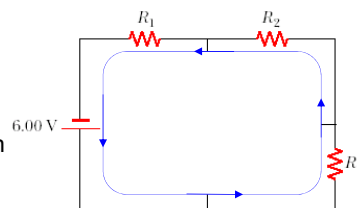


Resistors in Parallel



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Example. A 6.00-V battery supplies current to the circuit shown in figure. When the double-throw switch S is open, as shown in the figure, the current in the battery is 1.00 mA. When the switch is closed in position 1, the current in the battery is 1.20 mA. When the switch is closed in position 2, the current in the battery is 2.00 mA. Find the resistances R_1 , R_2 , and R_3 .

**Answer:**

When S is open, R_1 , R_2 , R_3 are in series with the battery. Thus:

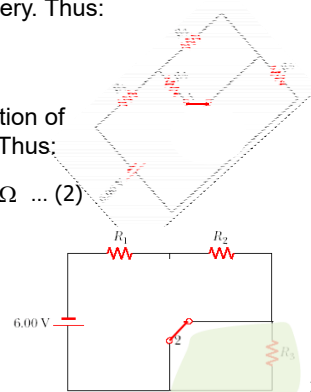
$$R_1 + R_2 + R_3 = \frac{\Delta V}{I} = \frac{6 \text{ V}}{1 \times 10^{-3} \text{ A}} = 6 \text{ k}\Omega \quad \dots (1)$$

When S is closed in position 1, the parallel combination of the two R_2 's is in series with R_1 , R_3 , and the battery. Thus:

$$R_1 + \frac{1}{\frac{1}{R_2} + \frac{1}{R_2}} + R_3 = R_1 + \frac{R_2}{2} + R_3 = \frac{6 \text{ V}}{1.2 \times 10^{-3} \text{ A}} = 5 \text{ k}\Omega \quad \dots (2)$$

When S is closed in position 2, R_1 and R_2 are in series with the battery. R_3 is shorted. Thus:

$$R_1 + R_2 = \frac{6 \text{ V}}{2 \times 10^{-3} \text{ A}} = 3 \text{ k}\Omega \quad \dots (3)$$

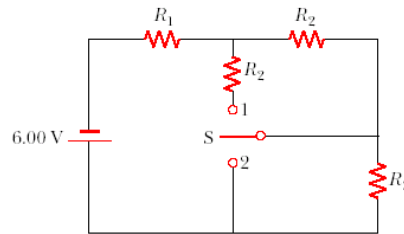


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$$R_1 + R_2 + R_3 = 6 \text{ k}\Omega \quad \dots (1)$$

$$R_1 + \frac{R_2}{2} + R_3 = 5 \text{ k}\Omega \quad \dots (2)$$

$$R_1 + R_2 = 3 \text{ k}\Omega \quad \dots (3)$$



The rest is algebra!!

From (1) and (3): $3 \text{ k}\Omega + R_3 = 6 \text{ k}\Omega \rightarrow R_3 = \boxed{3 \text{ k}\Omega}$

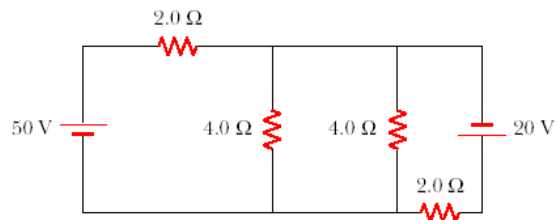
Subtract (2) from (1): $\frac{R_2}{2} = 1 \text{ k}\Omega \rightarrow R_2 = \boxed{2 \text{ k}\Omega}$

From (3): $R_1 + 2 \text{ k}\Omega = 3 \text{ k}\Omega \rightarrow R_1 = \boxed{1 \text{ k}\Omega}$

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Conservation of Charges: Whatever go into a junction must come out from the junction.

Conservation of Energy: The change in potential energy is 0 if charges come back to the start point on a loop.



Resistors in Series $R = R_1 + R_2 + R_3 + \dots$

Resistors in Parallel $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$

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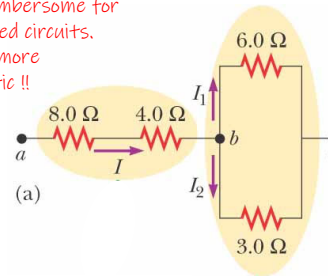
Combinations of Resistors

- The 8.0-Ω and 4.0-Ω resistors are in series and can be replaced with their equivalent, 12.0 Ω
- The 6.0-Ω and 3.0-Ω resistors are in parallel and can be replaced with their equivalent

$$\frac{1}{\frac{1}{6} + \frac{1}{3}} = \frac{1}{\left(\frac{3}{6}\right)} = 2.0 \, \Omega$$

- These equivalent resistances are in series and can be replaced with their equivalent resistance, 14.0 Ω

The use of this method will be cumbersome for complicated circuits. Let's be more systematic !!



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Kirchhoff's Rules

- Two rules, called **Kirchhoff's rules**, can be used to analyse complicated circuits

- Junction Rule (Conservation of Charge)

- The sum of the currents entering any junction must equal the sum of the currents leaving that junction

$$\Sigma I_{\text{in}} = \Sigma I_{\text{out}}$$

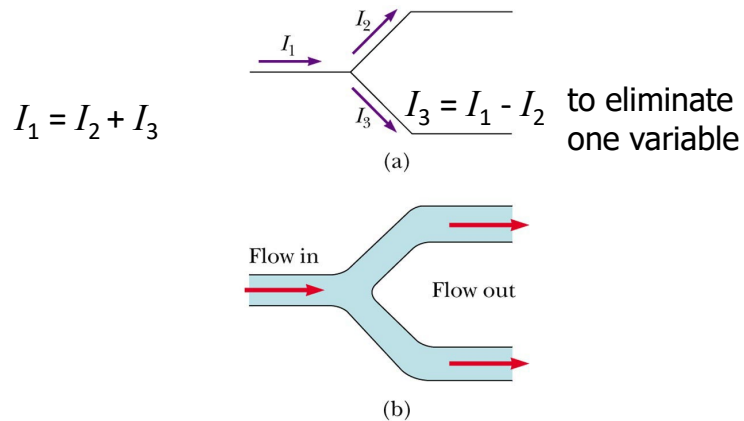
- Loop Rule (Conservation of Energy)

- The sum of the potential differences across all the elements around any closed circuit loop must be zero

$$\sum_{\text{closed loop}} \Delta V = 0$$

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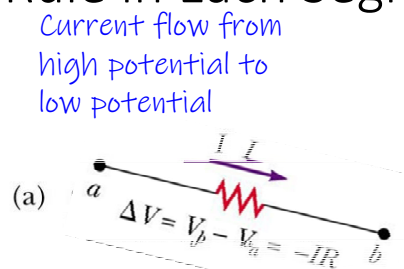
More about the Junction Rule



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More about the Loop Rule in Each Segment

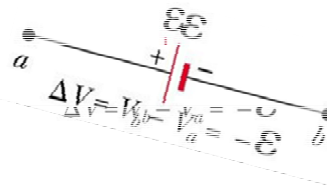
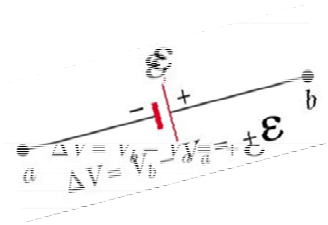
- From a to b in figure (a) the resistor is traversed in the direction of the current, the potential across the resistor is $-IR$



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More about the Loop Rule in Each Segment

- From a to b, in figure (c) the source of emf is traversed in the direction from $-$ to $+$, (c) and the change in the electric potential is $+\epsilon$
- From a to b in figure (d) the source of emf is traversed in the direction from $+$ to $-$, (d) and the change in the electric potential is $-\epsilon$



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Junction Equations from Kirchhoff's Rules

- Use the junction rule as often as needed, so long as each time you write an equation, you include in it a current that has not been used in a previous junction rule equation
 - In general, the number of times the junction rule can be used is one fewer than the number of junction points in the circuit

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Loop Equations from Kirchhoff's Rules

- The loop rule can be used as often as needed so long as a new circuit element (resistor or battery) or a new current appears in each new equation
- You need as many independent equations as you have unknowns

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Kirchhoff's Rules Equations, final

- In order to solve a particular circuit problem, the number of independent equations you need to obtain from the two rules equals the number of unknown currents
- Any capacitor acts as an open branch in a circuit
- The current in the branch containing the capacitor is zero under steady-state conditions

Not in PC1201 Syllabus!!

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Problem-Solving Hints – Kirchhoff's Rules

- Draw the circuit diagram and assign labels and symbols to all known and unknown quantities. Assign directions to the currents.
 - The direction is arbitrary, but you must adhere to the assigned directions when applying Kirchhoff's rules
- Apply the junction rule to any junction in the circuit that provides new relationships among the various currents

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Problem-Solving Hints, cont

- Apply the loop rule to as many loops as are needed to solve for the unknowns
 - To apply the loop rule, you must correctly identify the potential difference as you cross various elements
- Solve the equations simultaneously for the unknown quantities
 - If a current turns out to be a negative value, the magnitude of current will be still correct but the direction is opposite to what you have assigned

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Example. Calculate the current flow in each resistor shown in figure.

Answer:

Let the currents be as shown.

We apply Kirchhoff's rules to the diagram.

$$50 - 2I_1 - 2I_2 = 0 \dots (1)$$

$$20 - 2I_3 + 2I_2 = 0 \dots (2)$$

$$I_1 = I_2 + I_3 \dots (3)$$

These are algebra!!

Sub (3) to (1) $\rightarrow 50 - 2(I_2 + I_3) - 2I_2 = 0$

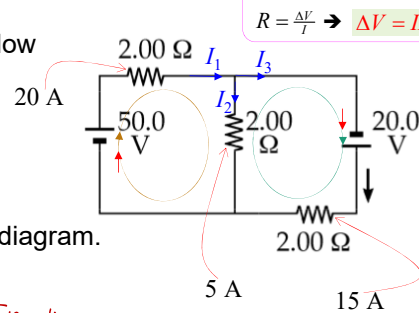
$$\rightarrow 50 - 4I_2 - 2I_3 = 0 \rightarrow -2I_3 = 4I_2 - 50 \dots (4)$$

Sub (4) to (2) $\rightarrow 20 + (4I_2 - 50) + 2I_2 = 0 \rightarrow 6I_2 - 30 = 0 \rightarrow I_2 = \boxed{5 \text{ A}}$

(1) $\rightarrow 2I_1 = 50 - 2I_2 = 50 - 2(5) = 40 \text{ A} \rightarrow I_1 = \boxed{20 \text{ A}}$

(3) $\rightarrow I_3 = I_1 - I_2 = 20 - 5 = \boxed{15 \text{ A}}$

Traditional approach presented in Textbook!!



$$R = \frac{\Delta V}{I} \rightarrow \Delta V = IR$$

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Simplified Approach I
have learnt from MIT
Professor Walter Lewin
from YouTube video:

Let the **partial** current in each cycle be as shown.

We apply Kirchhoff's rules to the diagram.

$$50 - 2I_1 + 2(-I_1 + I_2) = 0$$

$$\rightarrow 50 - 4I_1 + 2I_2 = 0 \dots (1)$$

$$20 - 2I_2 + 2(-I_2 + I_1) = 0$$

$$\rightarrow 20 - 4I_2 + 2I_1 = 0 \dots (2)$$

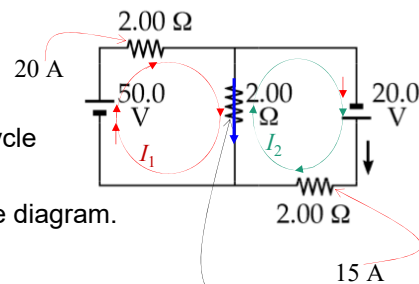
These are algebra!!

(2) $\times 2$: $40 - 8I_2 + 4I_1 = 0 \dots (3)$

(1) + (3): $90 - 6I_2 = 0 \rightarrow I_2 = \boxed{15 \text{ A}}$

(2): $20 - 4 \times 15 + 2I_1 = 0 \rightarrow I_1 = \boxed{20 \text{ A}}$

Simplified approach

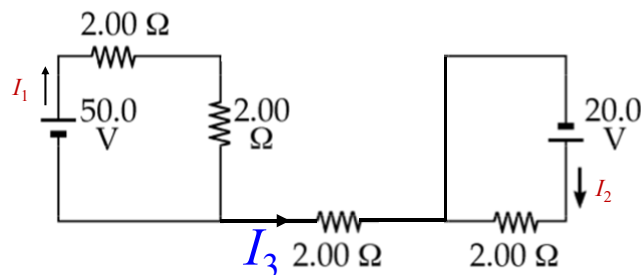


$$R = \frac{\Delta V}{I} \rightarrow \Delta V = IR$$

$$20 \downarrow - 15 \uparrow = 5 \text{ A}$$

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What will be the value of I_3 ?



Answer:

$I_3 = 0$ because there is no way to form a current loop if I_3 is non-zero in a circuit.
So this diagram includes 2 exclusive circuits.

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Example. Calculate the power delivered to each resistor shown in figure.

Simplified Answer:

Let the partial current in each cycle be as shown.

We apply Kirchhoff's rules to the diagram.

$$50 - 2I_1 + 4(-I_1 + I_2) = 0$$

$$\rightarrow 50 - 6I_1 + 4I_2 = 0 \rightarrow 25 - 3I_1 + 2I_2 = 0 \dots (1)$$

$$4(-I_2 + I_1) + 4(-I_2 + I_3) = 0 \rightarrow -2I_2 + I_1 + I_3 = 0 \dots (2)$$

$$20 - 2I_3 + 4(-I_3 + I_2) = 0$$

$$\rightarrow 20 - 6I_3 + 4I_2 = 0 \rightarrow 10 - 3I_3 + 2I_2 = 0 \dots (3)$$

$$(2) \rightarrow I_3 = 2I_2 - I_1 \dots (4)$$

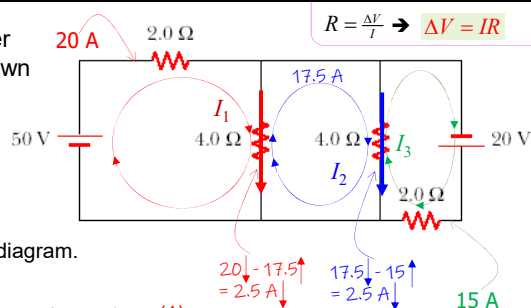
$$\text{Sub (4) to (3): } 10 - 3(2I_2 - I_1) + 2I_2 = 0 \rightarrow 10 - 4I_2 + 3I_1 = 0 \dots (5)$$

$$\text{Solve (1) and (5) by adding: } 35 - 2I_2 = 0 \rightarrow I_2 = 17.5 \text{ A}$$

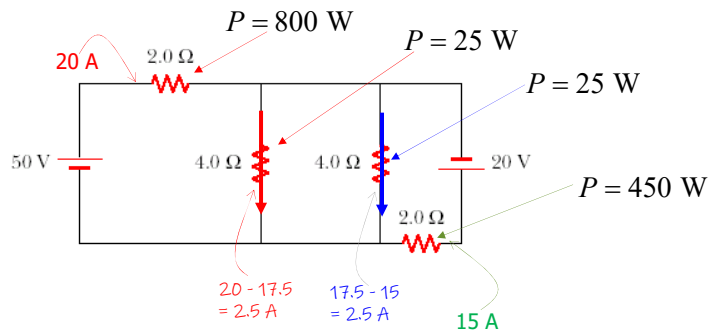
$$(1) \rightarrow 25 - 3I_1 + 2(17.5) = 0 \rightarrow I_1 = 20 \text{ A}$$

$$(3) \rightarrow 10 - 3I_3 + 2(17.5) = 0 \rightarrow I_3 = 15 \text{ A}$$

These are algebraic!



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To find the power delivered to each resistor, we apply $P = I^2 R$ to each resistor:

$$(2.00 \, \Omega): P = (20 \, \text{A})^2 (2 \, \Omega) = \boxed{800 \, \text{W}}$$

$$(4.00 \, \Omega): P = (2.5 \, \text{A})^2 (4.00 \, \Omega) = \boxed{25.0 \, \text{W}}$$

$$(2.00 \, \Omega): P = (15.0 \, \text{A})^2 (2.00 \, \Omega) = \boxed{450 \, \text{W}}$$

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To find the power delivered to each resistor, we apply $P = I^2 R$ to each resistor:

$$(2.00 \, \Omega): P = (20 \, \text{A})^2 (2 \, \Omega) = \boxed{800 \, \text{W}}$$

$$(4.00 \, \Omega): P = (2.5 \, \text{A})^2 (4.00 \, \Omega) = \boxed{25.0 \, \text{W}}$$

$$(2.00 \, \Omega): P = (15.0 \, \text{A})^2 (2.00 \, \Omega) = \boxed{450 \, \text{W}}$$

But energy must be conserved. In 1 second, the total energy delivered to the 4 resistors must be equal to the energy supplied by the 2 batteries!! This means that the total power received must equal to the total power supplied.

$$\begin{aligned} \text{Total power received by the 4 resistors} \\ = 800 \, \text{W} + 25 \, \text{W} + 25 \, \text{W} + 450 \, \text{W} = 1300 \, \text{W} \end{aligned}$$

$$\begin{aligned} \text{Total power } (P = I \times \Delta V) \text{ supplied by the 2 batteries} \\ = 20 \, \text{A} \times 50 \, \text{V} + 15 \, \text{A} \times 20 \, \text{V} = 1300 \, \text{W} \end{aligned}$$

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Example from Student.

Calculate the current flowing in each resistor shown in figure.

Simplified Answer:

Let the partial current in each cycle be as shown.

Apply Kirchhoff's rules to the left loop:

$$50 - 2I_1 + 4(-I_1 + I_2) = 0$$

$$\rightarrow 50 - 6I_1 + 4I_2 = 0 \rightarrow 25 - 3I_1 + 2I_2 = 0 \dots (1)$$

Apply Kirchhoff's rules to the middle loop:

$$4(-I_2 + I_1) + 4(-I_2 - I_3) = 0 \rightarrow -8I_2 + 4I_1 - 4I_3 = 0 \rightarrow I_1 - 2I_2 - I_3 = 0 \dots (2)$$

Apply Kirchhoff's rules to the right loop:

$$20 + 4(-I_3 - I_2) - 2I_3 = 0 \rightarrow 20 - 6I_3 - 4I_2 = 0 \rightarrow I_3 = \frac{10 - 2I_2}{3} \dots (3)$$

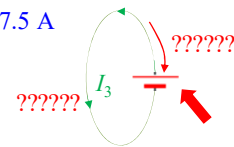
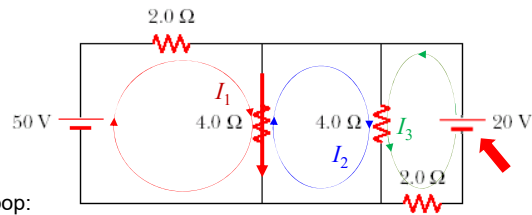
$$\text{Sub (3) to (2): } I_1 - 2I_2 - \frac{10 - 2I_2}{3} = 0 \rightarrow 3I_1 - 6I_2 - 10 + 2I_2 = 0 \rightarrow -3I_1 = -4I_2 - 10 \dots (4)$$

$$\text{Sub (4) to (1): } 25 - 4I_2 - 10 + 2I_2 = 0 \rightarrow 15 = 2I_2 \rightarrow I_2 = 7.5 \text{ A}$$

$$(1) \rightarrow 25 - 3I_1 + 2 \times 7.5 = 0 \rightarrow 40 = 3I_1 \rightarrow I_1 = 13.3333 \text{ A}$$

$$(3) \rightarrow I_3 = \frac{10 - 2 \times 7.5}{3} \rightarrow I_3 = -1.6667 \text{ A} \text{ ?????}$$

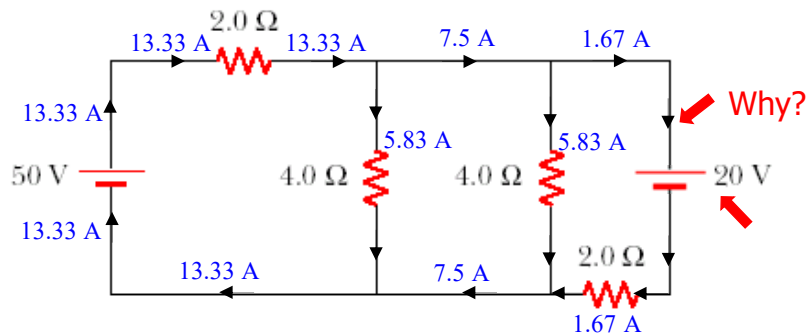
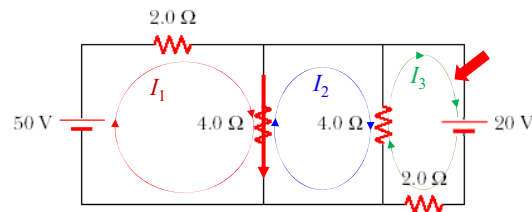
These are algebraic!

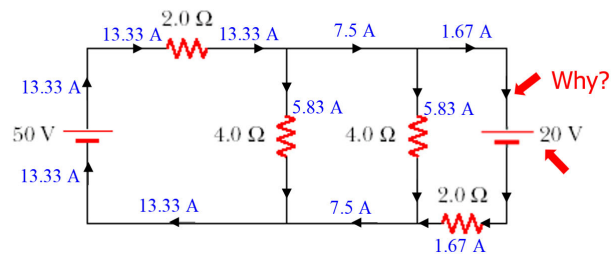


Example. Calculate the current flowing in each resistor shown in figure.

Simplified Answer:

$$I_1 = 13.3333 \text{ A (clockwise)} \quad I_2 = 7.5 \text{ A (clockwise)} \quad I_3 = 1.6667 \text{ A (clockwise)}$$





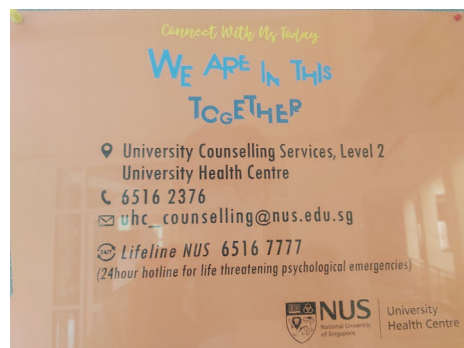
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Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

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- End -

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